



Dr. Ming-Zhe Chong



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Education and Work Background

- 2021.9 - 2026.6 **Peking University** (PKU, School of Electronics, Supervisor: Prof. Ming-Yao Xia)
Ph.D. Major: Electromagnetic Field and Microwave Technology
- 2025.5 - 2025.8 **City University of Hong Kong** (CityUHK, State Key Laboratory of Terahertz and Millimeter Waves)
Research Assistant
- 2017.9 - 2021.6 **Huazhong University of Science and Technology** (HUST, School of Optical and Electronic Information)
B.Eng. Major: Optoelectronic Information Science and Engineering
- 2019.9 - 2020.1 **Zhejiang University** (ZJU, School of Optical Science and Engineering) Exchange Student



Research Interests and Publications

- Research Interests:** Terahertz (THz), Metasurfaces, EM wave manipulation
 - 1. Diffractive optical neural networks:** Based on the diffraction propagation features between multiple layers of metasurfaces, diffractive network models are trained to achieve functions such as **non-reciprocal imaging**;
 - 2. Terahertz communication:** We design terahertz OAM mode-division-multiplexing wireless links, including mode conversion and filtering functions, and verify the large communication capacity of terahertz channels;
 - 3. Structured beam manipulation:** The metasurfaces are designed and fabricated, enabling the realization of various terahertz structured beams such as focusing arrays, OAM combs, and surface Bessel beams.
- Publications:** I have **7** SCI papers accepted or published as the (co-)first author; **1** paper published as the first author in an EI conference.
SCI papers:
 - Ming-Zhe Chong**, et al. "Orbital angular momentum processing via diffractive networks for terahertz mode-division-multiplexed wireless links." *Light: Science & Applications*, 2026, Accepted. (IF=23.4)
 - Ming-Zhe Chong**, et al. "Janus meta-imager: asymmetric image transmission and transformation enabled by diffractive neural networks." *Photonix*, 2025, 6(1): 60. (IF=19.1)
 - Ming-Zhe Chong**, et al. "Reconfigurable terahertz beam splitters enabled by inverse-designed meta-devices." *Photonics Research*, 2025, 13(8): 2328-2338. (IF=7.2)
 - Ming-Zhe Chong**, et al. "Generation of polarization-multiplexed terahertz orbital angular momentum combs via all-silicon metasurfaces." *Light: Advanced Manufacturing*, 2024, 5(3): 400-409. (IF=10.6)
 - Ming-Zhe Chong**, et al. "Spin-decoupled excitation and wavefront shaping of structured surface waves via on-chip terahertz metasurfaces." *Nanoscale*, 2023, 15(9): 4515-4522. (IF=5.1)
 - Ming-Zhe Chong**, et al. "Nonlinear modulation of terahertz waves based on a MAPbI₃/Gold/Si hybrid plasmon-induced transparency (PIT) metasurface." *Optical Materials*, 2022, 129: 112554. (IF=4.2)



Awards and Honours

- 2026 PKU **Outstanding Graduate**
- 2025 PKU **BYD Scholarship**
- 2025 PKU **Award for Scientific Research**
- 2024 PKU **Merit Student**
- 2024 PKU "Challenge Cup":
Special Prize of the Youth Science Award
- 2018 **The Chinese Mathematics Competition**
for College Students: **First Prize**



Other

- Skills:** 1. Proficient in **MATLAB** and **Python** programming;
2. Familiar with graphic design software,
such as **Adobe Illustrator**;
3. Proficient in electromagnetic simulation software,
such as **CST**;
4. Familiar with **terahertz experiment**,
be able to build and operate THz testing systems.
- English:** **IELTS** (Academic) **7.0**
- Hobbies:** Mountaineering, hiking, badminton, etc.